

Héber H. Arcolezi, Ph.D.

 <https://hharcolezi.github.io/>
 heber.hwang-arcolezi@etsmtl.ca

 Google Scholar
 @hharcolezi

 [heber-hwang-arcolezi](#)

Professional Experience

- Feb 2026 – Present  **Tenure-Track Assistant Professor**, ÉTS Montréal, Canada.
- Oct 2023 – Jan 2026  **Tenured Research Scientist (ISFP)**¹, Inria Grenoble, France.
- Feb 2022 – Sept 2023  **Postdoctoral Researcher**, Inria Saclay & École Polytechnique, France.

Education

- Jun 2019 – Jan 2022  **Ph.D. in Computer Science**, Université Bourgogne Franche-Comté (UBFC)
Thesis: *Production of Categorical Data Verifying Differential Privacy: Conception and Applications to Machine Learning* [link]
Advisors: Jean-François Couchot, Bechara Al Bouna, and Xiaokui Xiao
- Sep 2017 – Aug 2019  **M.Sc. in Electrical Engineering**, São Paulo State University (UNESP)
Thesis: *A Novel Robust and Intelligent Control Based Approach for Human Lower Limb Rehabilitation via Neuromuscular Electrical Stimulation* [link]
Advisor: Aparecido A. de Carvalho
- Jul 2012 – Aug 2017  **B.Eng. in Electrical Engineering**, Mato Grosso State University (UNEMAT)
Thesis: *Um Estudo Complementar ao Projeto de Controle PID no Caso do Pêndulo Invertido (in Portuguese)* [link]
Advisor: Rogério B. Quirino

Publications

Conference Proceedings

- H. H. Arcolezi**, “Private frequency estimation via residue number systems,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, 2026, pp. 35 518–35 526.  DOI: 10.1609/aaai.v40i42.40862
- H. H. Arcolezi** and S. Gambs, “Revisiting locally differentially private protocols: Towards better trade-offs in privacy, utility, and attack resistance,” in *2026 IEEE 42nd International Conference on Data Engineering (ICDE)*, 2026, pp. 294–307.  DOI: 10.1109/ICDE65706.2026.00029
- M. Babaei, Y. Wang, H. Lautraite, **H. H. Arcolezi**, U. Aivodji, and S. Gambs, “Quantifying the privacy of counterfactuals by leveraging membership inference attacks against synthetic data,” in *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency*, ser. FAccT ’26, New York, NY, USA: Association for Computing Machinery, 2026, pp. 4988–5022, ISBN: 9798400725968.  DOI: 10.1145/3805689.3812361
- R. G. Gonze, N. Fernandes, **H. H. Arcolezi**, C. Palamidessi, and N. Bielova, “Beyond Epsilon: A Principled QIF Framework for Local Differential Privacy,” in *2026 39th IEEE Computer Security Foundations Symposium (CSF)* CSF, Los Alamitos, CA, USA: IEEE Computer Society, Jul. 2026, pp. 483–496.  DOI: 10.1109/CSF68417.2026.00032
- A. A. Marreiras Neto, **H. H. Arcolezi**, and J. C. Machado, “Data poisoning in longitudinal local differential privacy: Attacks and analysis,” in *Data and Applications Security and Privacy XL*, Springer Nature Switzerland, 2026, pp. 124–141, ISBN: 9783032332608.  DOI: 10.1007/978-3-032-33260-8_7

¹Inria ISFP researchers have a light teaching load, ranging from approximately 32 to 64 hours per year. More information at <https://www.inria.fr/fr/inria-starting-faculty-position-isfp>.

- 6 A. K. Mishra, M. Cunche, and **H. H. Arcolezi**, “How tough is location anonymization? re-identifying 100k real-user trajectories in japan,” in *Proceedings of the ACM Asia Conference on Computer and Communications Security*, ser. ASIA CCS ’26, Bangalore, India: Association for Computing Machinery, 2026, pp. 1261–1273, ISBN: 9798400723568.  DOI: 10.1145/3779208.3805978
- 7 C. A. Pinzón, E. ElSalamouny, L. Massot, A. Miller, **H. H. Arcolezi**, and C. Palamidessi, “Estimating the true distribution of data collected with randomized response,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, 2026, pp. 35 751–35 758.  DOI: 10.1609/aaai.v40i42.40888
- 8 **H. H. Arcolezi**, M. Alishahi, A.-A. Bendoukha, and N. Kaaniche, “Fair play for individuals, foul play for groups? auditing anonymization’s impact on ml fairness,” in *ECAI 2025*, IOS Press, Oct. 2025.  DOI: 10.3233/faia250909
- 9 A.-A. Bendoukha, **H. H. Arcolezi**, N. Kaaniche, A. Boudguiga, R. Sirdey, and P.-E. Clet, “Fade: Federated aggregation with discrimination elimination,” in *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*, ser. FAccT ’25, Association for Computing Machinery, 2025, pp. 3182–3195, ISBN: 9798400714825.  DOI: 10.1145/3715275.3732203
- 10 C. Eichler, N. Champeil, N. Anciaux, A. Bensamoun, **H. H. Arcolezi**, and J. M. De Fuentes, “Nob-MIAs: Non-biased membership inference attacks assessment on large language models with ex-post dataset construction,” in *Web Information Systems Engineering – WISE 2024*, Singapore: Springer Nature Singapore, 2025, pp. 441–456.  DOI: 10.1007/978-981-96-0570-5_32
- 11 H. Zhang, A. K. Mishra, and **H. H. Arcolezi**, “Demo: Exploring utility and attackability trade-offs in local differential privacy,” in *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*, ser. CCS ’25, Taipei, Taiwan: Association for Computing Machinery, 2025, pp. 4728–4730.  DOI: 10.1145/3719027.3760706
- 12 R. Binkyte, C. A. Pinzón, S. Lestyán, K. Jung, **H. H. Arcolezi**, and C. Palamidessi, “Causal discovery under local privacy,” in *Proceedings of the Third Conference on Causal Learning and Reasoning*, F. Locatello and V. Didelez, Eds., ser. Proceedings of Machine Learning Research, vol. 236, PMLR, Jan. 2024, pp. 325–383.  URL: <https://proceedings.mlr.press/v236/binkyte24a.html>
- 13 K. Makhlof, T. Stefanović, **H. H. Arcolezi**, and C. Palamidessi, “A systematic and formal study of the impact of local differential privacy on fairness: Preliminary results,” in *2024 IEEE 37th Computer Security Foundations Symposium (CSF)*, 2024, pp. 1–16.  DOI: 10.1109/CSF61375.2024.00039
- 14 **H. H. Arcolezi**, S. Cerna, and C. Palamidessi, “On the utility gain of iterative bayesian update for locally differentially private mechanisms,” in *Data and Applications Security and Privacy XXXVII*, Springer Nature Switzerland, 2023, pp. 165–183, ISBN: 9783031375866.  DOI: 10.1007/978-3-031-37586-6_11
- 15 **H. H. Arcolezi**, K. Makhlof, and C. Palamidessi, “(local) differential privacy has no disparate impact on fairness,” in *Data and Applications Security and Privacy XXXVII*, V. Atluri and A. L. Ferrara, Eds., Cham: Springer Nature Switzerland, 2023, pp. 3–21.  DOI: 10.1007/978-3-031-37586-6_1
- 16 **H. H. Arcolezi**, C. A. Pinzón, C. Palamidessi, and S. Gambs, “Frequency estimation of evolving data under local differential privacy,” in *Proceedings of the 26th International Conference on Extending Database Technology, EDBT 2023, Ioannina, Greece, March 28 - March 31, 2023*, OpenProceedings.org, 2023, pp. 512–525.  DOI: 10.48786/EDBT.2023.44
- 17 **H. H. Arcolezi**, J.-F. Couchot, S. Gambs, C. Palamidessi, and M. Zolfaghari, “Multi-freq-ldpy: Multiple frequency estimation under local differential privacy in python,” in *Computer Security – ESORICS 2022*, V. Atluri, R. Di Pietro, C. D. Jensen, and W. Meng, Eds., Cham: Springer Nature Switzerland, 2022, pp. 770–775.  DOI: 10.1007/978-3-031-17143-7_40
- 18 **H. H. Arcolezi**, J.-F. Couchot, B. Al Bouna, and X. Xiao, “Random sampling plus fake data: Multidimensional frequency estimates with local differential privacy,” in *Proceedings of the 30th ACM International Conference on Information & Knowledge Management*, ACM, Oct. 2021, pp. 47–57.  DOI: 10.1145/3459637.3482467

- 19 **H. H. Arcolezi**, J.-F. Couchot, B. A. Bouna, and X. Xiao, "Longitudinal collection and analysis of mobile phone data with local differential privacy," in *IFIP International Summer School on Privacy and Identity Management*, Springer International Publishing, 2021, pp. 40–57, ISBN: 9783030724658.  DOI: 10.1007/978-3-030-72465-8_3
- 20 **H. H. Arcolezi**, J.-F. Couchot, O. Baala, J.-M. Contet, B. Al Bouna, and X. Xiao, "Mobility modeling through mobile data: Generating an optimized and open dataset respecting privacy," in *2020 International Wireless Communications and Mobile Computing (IWCMC)*, 2020, pp. 1689–1694.  DOI: 10.1109/IWCMC48107.2020.9148138
- 21 S. Cerna, C. Guyeux, **H. H. Arcolezi**, R. Couturier, and G. Royer, "A comparison of lstm and xgboost for predicting firemen interventions," in *Trends and Innovations in Information Systems and Technologies*, Springer International Publishing, 2020, pp. 424–434, ISBN: 9783030456917.  DOI: 10.1007/978-3-030-45691-7_39
- 22 S. Cerna, C. Guyeux, **H. H. Arcolezi**, and G. Royer, "Boosting methods for predicting firemen interventions," in *2020 11th International Conference on Information and Communication Systems (ICICS)*, 2020, pp. 001–006.  DOI: 10.1109/ICICS49469.2020.239488
- 23 **H. H. Arcolezi**, W. R. B. M. Nunes, S. L. C. Ñahuis, M. A. A. Sanches, M. C. M. Teixeira, and A. A. de Carvalho, "A rise-based controller fine-tuned by an improved genetic algorithm for human lower limb rehabilitation via neuromuscular electrical stimulation," in *2019 6th International Conference on Control, Decision and Information Technologies (CoDIT)*, 2019, pp. 1197–1202.  DOI: 10.1109/CoDIT.2019.8820357
- 24 S. L. C. Ñahuis, C. Guyeux, **H. H. Arcolezi**, R. Couturier, G. Royer, and A. D. P. Lotufo, "Long short-term memory for predicting firemen interventions," in *2019 6th International Conference on Control, Decision and Information Technologies (CoDIT)*, 2019, pp. 1132–1137.  DOI: 10.1109/CoDIT.2019.8820671

Journal Articles

- 1 V. G. Angelozzi and **H. H. Arcolezi**, "Where to intervene? benchmarking fairness-aware learning on differentially private synthetic tabular data," *Proceedings on Privacy Enhancing Technologies*, vol. 2026, no. 4, pp. 53–82, Oct. 2026, ISSN: 2299-0984.  DOI: 10.56553/popets-2026-0110
- 2 P. Guerra-Balboa, A. Sauer, **H. H. Arcolezi**, and T. Strufe, "Understanding disclosure risk in differential privacy with applications to noise calibration and auditing," *Proceedings of the VLDB Endowment*, vol. 19, no. 7, pp. 1558–1571, Mar. 2026, ISSN: 2150-8097.  DOI: 10.14778/3801059.3801069
- 3 **H. H. Arcolezi**, K. Makhlof, and C. Palamidessi, "Group fairness under obfuscated sensitive information," *Journal of Computer Security*, vol. 33, no. 4, pp. 239–260, May 2025, ISSN: 1875-8924.  DOI: 10.1177/0926227x251330212
- 4 **H. H. Arcolezi** and S. Gambs, "Revealing the true cost of locally differentially private protocols: An auditing perspective," *Proceedings on Privacy Enhancing Technologies*, vol. 2024, no. 4, pp. 123–141, 2024.  DOI: 10.56553/popets-2024-0110
- 5 K. Makhlof, **H. H. Arcolezi**, S. Zhioua, G. B. Brahim, and C. Palamidessi, "On the impact of multi-dimensional local differential privacy on fairness," *Data Mining and Knowledge Discovery*, May 2024, ISSN: 1573-756X.  DOI: 10.1007/s10618-024-01031-0
- 6 **H. H. Arcolezi**, S. Gambs, J.-F. Couchot, and C. Palamidessi, "On the risks of collecting multidimensional data under local differential privacy," *Proc. VLDB Endow.*, vol. 16, no. 5, pp. 1126–1139, Jan. 2023, ISSN: 2150-8097.  DOI: 10.14778/3579075.3579086
- 7 **H. H. Arcolezi**, S. Cerna, J.-F. Couchot, C. Guyeux, and A. Makhoul, "Privacy-preserving prediction of victim's mortality and their need for transportation to health facilities," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 8, pp. 5592–5599, 2022.  DOI: 10.1109/TII.2021.3123588

- 8 **H. H. Arcolezi**, J.-F. Couchot, B. A. Bouna, and X. Xiao, "Improving the utility of locally differentially private protocols for longitudinal and multidimensional frequency estimates," *Digital Communications and Networks*, Jul. 2022.  DOI: 10.1016/j.dcan.2022.07.003
- 9 **H. H. Arcolezi**, J.-F. Couchot, D. Renaud, B. Al Bouna, and X. Xiao, "Differentially private multivariate time series forecasting of aggregated human mobility with deep learning: Input or gradient perturbation?" *Neural Computing and Applications*, vol. 34, no. 16, pp. 13 355–13 369, Jun. 2022, ISSN: 1433-3058.  DOI: 10.1007/s00521-022-07393-0
- 10 **H. H. Arcolezi**, S. Cerna, C. Guyeux, and J.-F. Couchot, "Preserving geo-indistinguishability of the emergency scene to predict ambulance response time," *Mathematical and Computational Applications*, vol. 26, no. 3, p. 56, Aug. 2021, ISSN: 2297-8747.  DOI: 10.3390/mca26030056
- 11 **H. H. Arcolezi** et al., "Rise controller tuning and system identification through machine learning for human lower limb rehabilitation via neuromuscular electrical stimulation," *Engineering Applications of Artificial Intelligence*, vol. 102, p. 104 294, Jun. 2021, ISSN: 0952-1976.  DOI: 10.1016/j.engappai.2021.104294
- 12 S. Cerna, **H. H. Arcolezi**, C. Guyeux, G. Royer-Fey, and C. Chevallier, "Machine learning-based forecasting of firemen ambulances' turnaround time in hospitals, considering the covid-19 impact," *Applied Soft Computing*, vol. 109, p. 107 561, Sep. 2021, ISSN: 1568-4946.  DOI: 10.1016/j.asoc.2021.107561
- 13 **H. H. Arcolezi** et al., "Forecasting the number of firefighter interventions per region with local-differential-privacy-based data," *Computers & Security*, vol. 96, p. 101 888, Sep. 2020, ISSN: 0167-4048.  DOI: 10.1016/j.cose.2020.101888
- 14 **H. H. Arcolezi** et al., "Identifying the knee joint angular position under neuromuscular electrical stimulation via long short-term memory neural networks," *Research on Biomedical Engineering*, vol. 36, no. 4, pp. 511–526, Sep. 2020, ISSN: 2446-4740.  DOI: 10.1007/s42600-020-00089-1

Grants

2026-04-01	 Discovery Grant: 192,500\$ Title: <i>Locally Private Representation Learning for Responsible Artificial Intelligence</i> Role: Principal and Unique Investigator
2025-04-01	 Chair Program: 500,000€ Title: <i>Responsible AI: Design, Regulation, and Conformity (RAIDAC+)</i> Role: Co-Chair with Prof. Theodore Christakis
2024-07-01	 ANR AAPG 2024 – JCJC (Young Researcher)²: 347,310€ Title: <i>Aligning Privacy, Utility, and Fairness for Responsible AI (AI-PULSE)</i> Role: Principal Investigator
2024-02-16	 Inria Associated Team³: 32,000€ Title: <i>Algorithmic Auditing of Privacy and Fairness (AUDIT-PAIR)</i> Role: Principal Investigator (Inria side)
2023-01-01	 ANR AAPG 2023 – PRCE (Academic and Industry Collaboration): 338,000€ Title: <i>Making PostgreSQL Differentially Private for Transparent AI (DIFPRIPOS)</i> Role: Member
2024-01-01	 Open call to sustain the development and promotion of AI: 10,000€ Title: <i>Exploring the Interplay of Differential Privacy and Fairness in ML</i> Role: Principal Investigator

²Appel à Projets Générique de l'Agence Nationale de la Recherche (ANR).

³This program funds mobility between Inria and an international research group for 3 years.

Awards

- 2025  **Notable Reviewer Award** at USENIX Security 2025
- 2024  **Best Reviewer Award** at PETS 2024
- 2023  **Best Paper Award** at DBSec 2023 for the paper “(Local) differential privacy has no disparate impact on fairness”
- 2021  **Ph.D. Student Mobility Grant (2,300€)** from the University Bourgogne Franche-Comté (UBFC) to visit the Université du Québec à Montréal (UQAM)
- 2013  **UNEMAT Scholarship (19,200R\$ – 400R\$/month)** to be a collaborator in the Formation of Cooperative Cells (FOCCO) program for 4 years (2013 – 2016)

Teaching

- Summer 2026  **Data Privacy**
Engineer Degree, 39h, ÉTS Montréal
- Winter 2025  **Database Management Systems**
Professional Bachelor in Networks and Telecommunication, 8h, Université Grenoble Alpes
-  **Python Programming Fundamentals**
Professional Bachelor in Networks and Telecommunication, 10h, Université Grenoble Alpes
- Spring 2024  **Database Management Systems**
Professional Bachelor in Networks and Telecommunication, 27h, Université Grenoble Alpes
- Spring 2023  **Introduction to Computer Science with Java**
Engineer degree and Bachelor, 40h, École Polytechnique
- Spring 2022  **Introduction to Computer Science with Java**
Engineer degree and Bachelor, 40h, École Polytechnique
- Spring 2021  **Privacy for IoT**
Master Internet of Things (M1), 20h, Université Bourgogne Franche-Comté
- Winter 2020  **Privacy for IoT**
Master Internet of Things (M1), 20h, Université Bourgogne Franche-Comté

Students

- 2025 - Present  **Marwa El Kamil**, INSA de Lyon & ÉTS Montréal
Type: PhD
Project Title: *Being exposed only to the risks I am willing to take*
Co-Supervisor: Ulrich Aïvodji, Antoine Boutet
-  **Zhan Xu**, Université Grenoble Alpes (UGA) & ÉTS Montréal
Type: PhD
Project Title: *Private and Fair Machine Learning*
Co-Supervisor: Ulrich Aïvodji
-  **Imen Bajar**, INSA de Lyon & ÉTS Montréal
Type: PhD
Project Title: *Assessment and reduction of re-identification risk from internal brain tissues of brain MRIs*
Co-Supervisor: Mohamed Maouche, Carole Frindel, Antoine Boutet

Students (continued)

- 2025  **Vinícius Gabriel Angelozzi**, Université Grenoble Alpes (UGA)
Type: MEng
Project Title: *Benchmarking Fairness-Aware Learning on Differentially Private Synthetic Tabular Data*
-  **Annika Sauer**, Karlsruhe Institute of Technology (KIT) & Université Grenoble Alpes (UGA)
Type: MEng
Project Title: *Measurement of Attack Resilience of Differential Privacy*
Co-Supervisor: Patricia Guerra-Balboa, Thorsten Strufe
-  **Mohammed Amine Benrahmoune**, Université Grenoble Alpes (UGA)
Type: Bachelor
Project Title: *Backdooring Machine Learning Models through Differentially Private Noise*
Co-Supervisor: Antoine Boutet
- 2022 - 2024  **Karima Makhoulf**, Institut Polytechnique de Paris (IPP)
Type: PhD
Project Title: *Advancing Ethical and Responsible AI: Exploring Fairness, Privacy, and Explainability through Causal Perspectives*
Co-Supervisor: Catuscia Palamidessi

Services

Program Committee

- 2027  VLDB, PETS
- 2026  CCS, AAI
- 2025  CCS, PETS, USENIX Security, ICLR
- 2024  CCS, PETS, IJCAI, FAccT, SAC, ICLR
- 2023  NeurIPS, FAccT, ECML PKDD, CCS, ICLR
- 2022  ECML PKDD

Conference & Workshop Organization

- 2026  7th Summer school on deep learning for medical imaging (DLMI 2026), Canada
- 2024  14th Atelier sur la Protection de la Vie Privée (APVP 2024)⁴, France
- 2023  13th Atelier sur la Protection de la Vie Privée (APVP 2023), France
- 2022  1st Comète Workshop on Ethical AI, France
- 2017  II Semana da Animação, Modelagem e Automação, Brazil
- 2014  I Semana da Faculdade de Ciencias Exatas, Brazil

Talks

- 2026  **Keynote Speaker:** *“Privacy Protection in Medical Imaging”* at 7th Summer School on Deep Learning for Medical Imaging (DLMI 2026)
- 2025  **Keynote Speaker:** *“Interplay of privacy and fairness in machine learning”* at 2nd Unite! Research School (URS 2025)
-  **Keynote Speaker:** *“Intersections of Fairness and Privacy: A Local Differential Privacy Perspective”* at 9th GDR RSD/ASF Winter School on Distributed Systems & Networks 2025

⁴Translates to “Workshop on Privacy”. This is the national French workshop from the privacy-preserving community GT-PVP. More info at <https://www.univ-orleans.fr/lifo/evenements/GT-PVP/>.

Talks (continued)

- 2024  **Invited Tutorial:** “Securing Data with Local Differential Privacy: Concepts, Protocols, and Practical Applications” at Selected Areas in Cryptography Summer School 2024
- 2023  **Invited Talk:** “Locally differentially private protocols for frequency estimation of longitudinal data” at Groupe de travail Protection de la Vie Privée (GT-PVP)
- 2022  **Invited Tutorial:** “A Brief Introduction to Local Differential Privacy” at SYSTOPIA Lab
-  **Invited Talk:** “Data anonymization and Artificial Intelligence Models (in Portuguese)” at Hospital Risoleta Tolentino Neves (HRTN)
- 2021  **Invited Talk:** “Improving Utility and Privacy in Multidimensional Frequency Estimates Under Local Differential Privacy” at Université du Québec à Montréal (LATECE Seminar)

Skills

- Languages  Strong reading, writing and speaking for English, French, Spanish, and Portuguese
- Coding  Python, Java, Matlab & Simulink, Visual Basic, LaTeX
- Databases  MySQL, PostgreSQL
- Misc.  Academic research, teaching, supervising, consultation, and publishing